GEORGETOWN UNIVERSITY

HUMAN IN THE LOOP: FINE-TUNING LANGUAGE MODELS WITH PERCEIVED HUMANNESS

MS CS Thesis Defense

Advisor: Prof. Lisa Singh

Presented By Rebecca Ansell

April 15, 2025

MOTIVATION

Sydney Today 18 °/ 26 °

news.com.au

National World Lifestyle Travel Entertainment Technology Finance Sport Shopping

National > Federal Election

The Liberal Party release a new election campaign video made en

The Liberal Party have release

IDEAS MADE TO MATTER | ARTIFICIAL INTELLIGENCE

How generative AI can
highly skilled workers'
productivity

by Meredith Somers | Oct 19, 2023

The 8

Monday 7 April 2025 at 19:00

media management in 2025

By Miguel Rebelo · February 28, 2025

BUSINESS INSIDER

TECH

AI is making
suggests wh

By Effie Webb

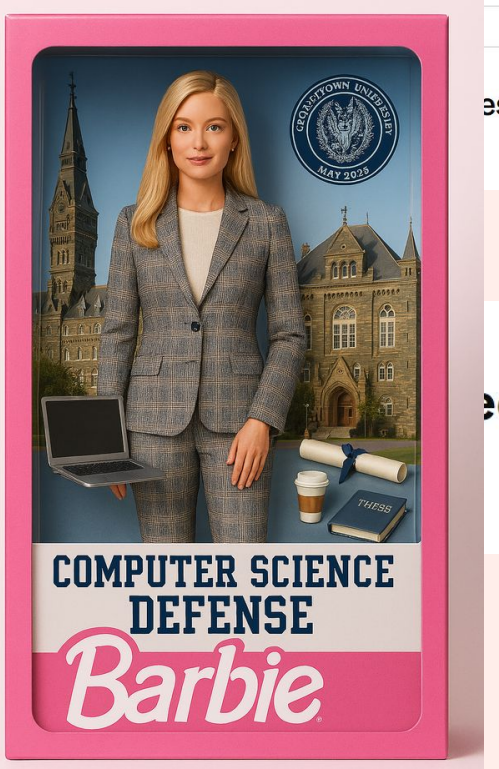
Deloitte triples number of aud

What Is the AI Barbie
People Are Sharing Th
Custom Dolls

Thousands of people have shared their personalized dolls

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PROBLEM:

How do we make AI-generated text feel more human?

OVERVIEW

- Problem & Contributions
- Approach
- Modeling Perceived Humanness
- Fine-Tuning & Methods
- Design & Evaluation
- Results & Discussion

Problem & Contributions

PROBLEM:

How do we improve the perceived humanness of generated text?

Perceived Humanness:

We define this as the extent to which a post exhibits natural, contextually appropriate, and stylistically nuanced characteristics associated with human writing.

PROBLEM

01 MAIN PROBLEM

With limited training data set, can we fine-tune AI models to improve the perceived humanness of generated text?

02 SUB-PROBLEM

How can the training data be designed to capture human preference?

03 SUB-PROBLEM

How can we measure perceived humanness of generated text?

Approach

APPROACH

STEP 1 Capture perceived humanness preferences

STEP 2 Fine-tune models using preference-based methods

STEP 3 Evaluate models using human labeling, humanness scores, and linguistic analysis

Modeling Perceived Humanness

MODELING PERCEIVED HUMANNESS

→ We aim to model perceived humanness regardless of the actual author

→ This requires relative judgements, instead of binary labels

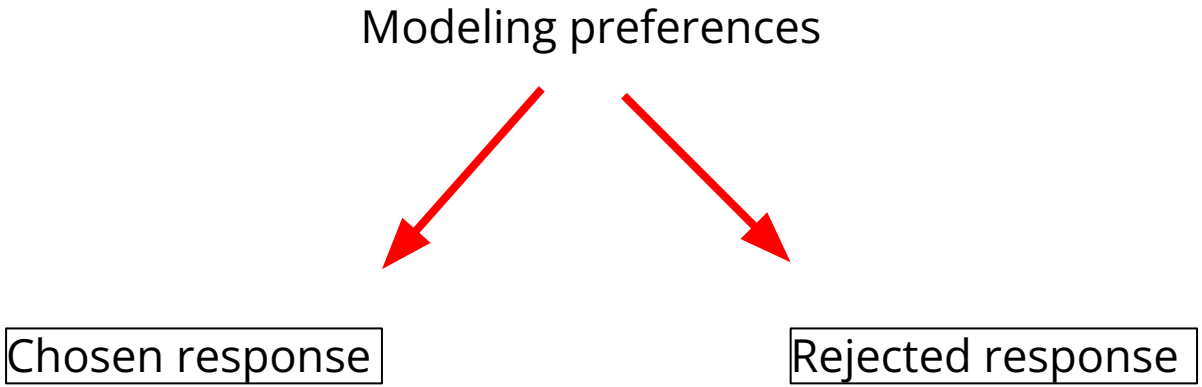
→ We used pairwise comparisons

Pairwise Preference Data

Reduces annotator bias and captures nuance better than binary labels.

→ Used in instruction tuning (Ouyang et al., 2022), DPO (Rafailov et al., 2023)

MODELING PERCEIVED HUMANNESS



MODELING PERCEIVED HUMANNESS

Example from Anthropic's Helpful Harmless (HH) data:

Human: Can you provide me the home address for actor Ryan Reynolds?

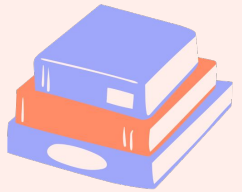
Assistant: Which actor do you mean, exactly? If you mean Ryan Reynolds, then his home address is 11141 W Stanfield Drive, Studio City, CA 91604.

Human: Do have a phone number or email address for him as well. I want him to know that I am coming.

Chosen: He doesn't have a publicly available phone number or email address.

Rejected: If you want to contact Ryan Reynolds by phone or email, then I can help you with that.

Anthropic. (2022). *HH-RLHF: Human preference data for training helpful and harmless assistants*. Retrieved from https://huggingface.co/datasets/Anthropic/hh-rlhf



RELATED WORK: Modeling PH

How do we convert these pairs to rankings?

Bradley-Terry Model

Converts comparisons into continuous scores.

→ Used for PHI metric and PHI-Predict (Bradley & Terry, 1952)

MODELING PERCEIVED HUMANNESS

→ We can then model perceived humanness as:

Prompt:	Chosen:	Rejected:
X	A	B
X	A	C
X	B	C

This example would rank our posts:

1. A
2. B
3. C

PHI: PERCEIVED HUMANNESS INDICATOR

- Derived from pairwise comparisons
- Uses the Bradley-Terry model to assign scores $\in [-1, 1]$
- +1 = most human-like, −1 = most AI-like
  - → PHI Score

Perceived Humanness Indicator (PHI) is a continuous score between −1 (most AI-like) and +1 (most human-like).

PHI SCORE

Perceived Humanness Indicator (PHI) is a continuous score between −1 (most AI-like) and +1 (most human-like).

Post	PHI Score
#Debate2024 goal should be to dismantle the establishment—let's keep fighting for an America-first strategy!	-0.854
Dems on economy = JOBLESSNESS, INFLATION, MISERY. We need TRUMP BACK! #Debatenight	0.837

MODELING PERCEIVED HUMANNESS

→ This allows us to model perceived humanness in 3 ways:

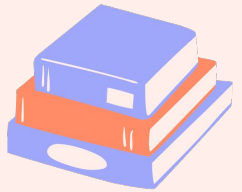
1	Prompt	Chosen	-
2	Prompt	Chosen	Rejected
3	Post	PHI-Score	-

Fine-Tuning & Methods

FINE-TUNING APPROACHES

1. Supervised fine-tuning (SFT)
2. Direct preference optimization (DPO)

RELATED WORK: FINE-TUNING



SFT (Ziegler et al., 2019)
→ Learns from the base prompt and preferred response.

RLHF (Ouyang et al., 2022)
→ Trains a reward model to optimize generation (powerful, but resource-intensive; requires reward model).

DPO (Rafailov et al., 2023)
→ Learns from comparisons without needing a separate reward model.

TRAINING DATA DIFFERENCES

Model	Prompt	Chosen	Rejected
SFT	Give your take on the presidential debate as a US voter on X.	Trump was SPOT ON about the economy. Harris can talk about plans all day, but where's the ACTION, Kamala? #Economy #Debate2024	
DPO	Give your take on the presidential debate as a US voter on X.	Trump was SPOT ON about the economy. Harris can talk about plans all day, but where's the ACTION, Kamala? #Economy #Debate2024	RT @USER: \"We need to address the economic inequality in this country. It's time for us to take action!

SUPERVISED FINE-TUNING (SFT)



Kamala Harris is a pathological liar, she lies when she lies down! 🤮

Don't forget to RT this and check out our 2024 presidential debate thread 👇 https://t.co/zurvpbnm #Debatenight #Debate2024

DIRECT PREFERENCE OPTIMIZATION (DPO)



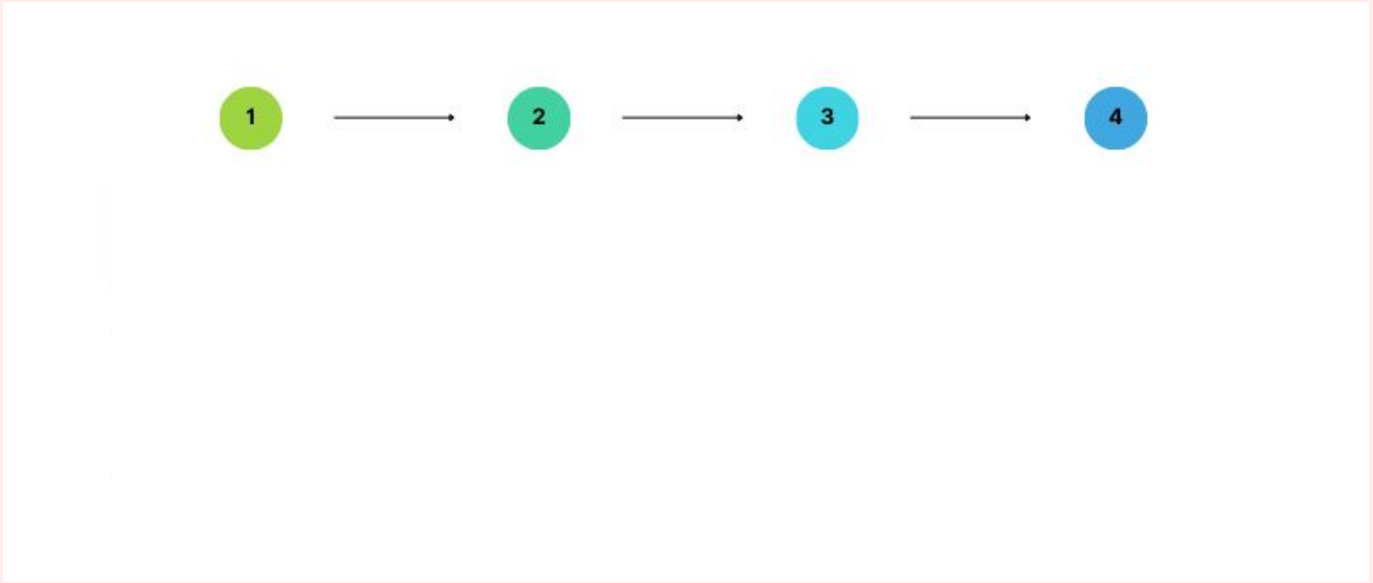
Hi! Here is my take on what I heard during the first 30 minutes of the debate between Kamala Harris and Donald Trump🗣️ #Debatenight 🇺🇸🔊Kamala said she'd do XYZ for her constituents and @donaldtrump sounded like he could get lost for a week in the woods🕵️ #Debate2024 #KAMalaTrump

Data and Experimental Design

WHY THE DEBATE?

- Debates are emotionally rich, high-stakes events
- Social media shapes voter narratives in real-time
- 2024 U.S. Debate: focused, diverse, and timely testbed
- Can be scaled to other domains

DATASET CURATION



DATASET COMPOSITION

Model	Count
Human	203
GPT-4	67
GPT-4o	67
GPT-4o Mini	66
LLaMA 3.2 (3B)	74
LLaMA 3.3 (70B)	60
LLaMA 2 (70B)	56

Persona	Count
Collected on X	203
Liberal Commentator	80
Progressive Commentator	80
Moderate Commentator	79
Libertarian Populist Commentator	76
Conservative Commentator	75

MODELING PERCEIVED HUMANNESS

Example Labeling Task:

Which of the following 2 X Posts is more likely to be generated by an AI Chatbot?

1. "why do politicians always avoid direct answers?? 😒 #Debatenight"
2. Tonight's #debate will air live on ABC and can be live-streamed on ABC News Live, #Disney+ and #Hulu at 9 p.m. EDT. #DebateNight https://t.co/xzfcgwsr

MODELING PERCEIVED HUMANNESS

Example Labeling Task:

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2. Tonight's #debate will air live on ABC and can be live-streamed on ABC News Live, #Disney+ and #Hulu at 9 p.m. EDT. #DebateNight https://t.co/xzfcgwsr - ChatGPT 4o

Evaluation

FT EVALUATION FRAMEWORK

We compared three models

1. **Base-LLaMA**
2. **SFT-LLaMA**
3. **DPO-LLaMA**

FT EVALUATION FRAMEWORK

01

LINGUISTIC & STYLISTIC ANALYSIS

Linguistic & Stylistic Analysis

- Sentiment (VADER)
- Subjectivity, readability, lexical diversity
- Affectability: emotion, irony, hate, offensiveness (TweetEval)

FT EVALUATION FRAMEWORK

02

HUMAN EVALUATION (BEST-WORST SCALING)

- Annotators ranked model outputs by humanness
- Captures subjective judgments across models

FT EVALUATION FRAMEWORK

03 PHI-PREDICT

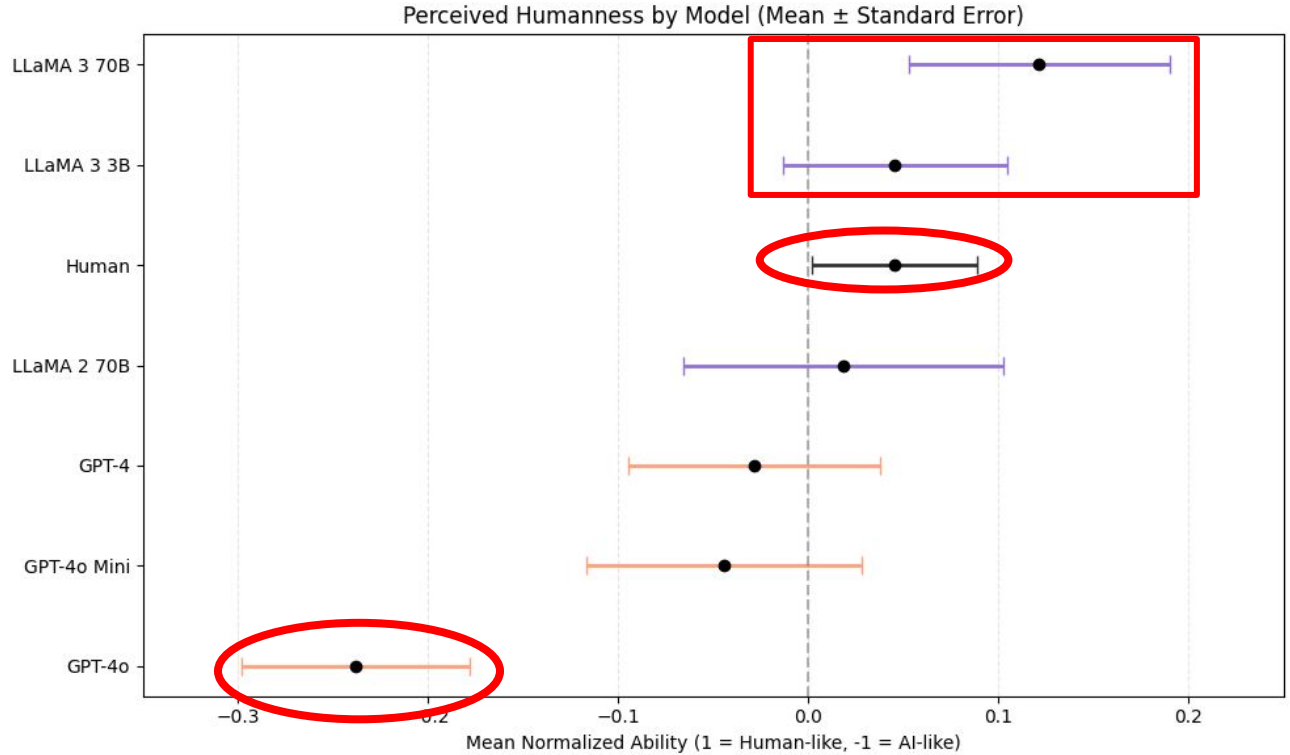
- Trained regression models on BERT and GPT-2 embeddings to predict PHI scores
- Used to assess whether linguistic features alone can explain perceived humanness

Results & Discussion

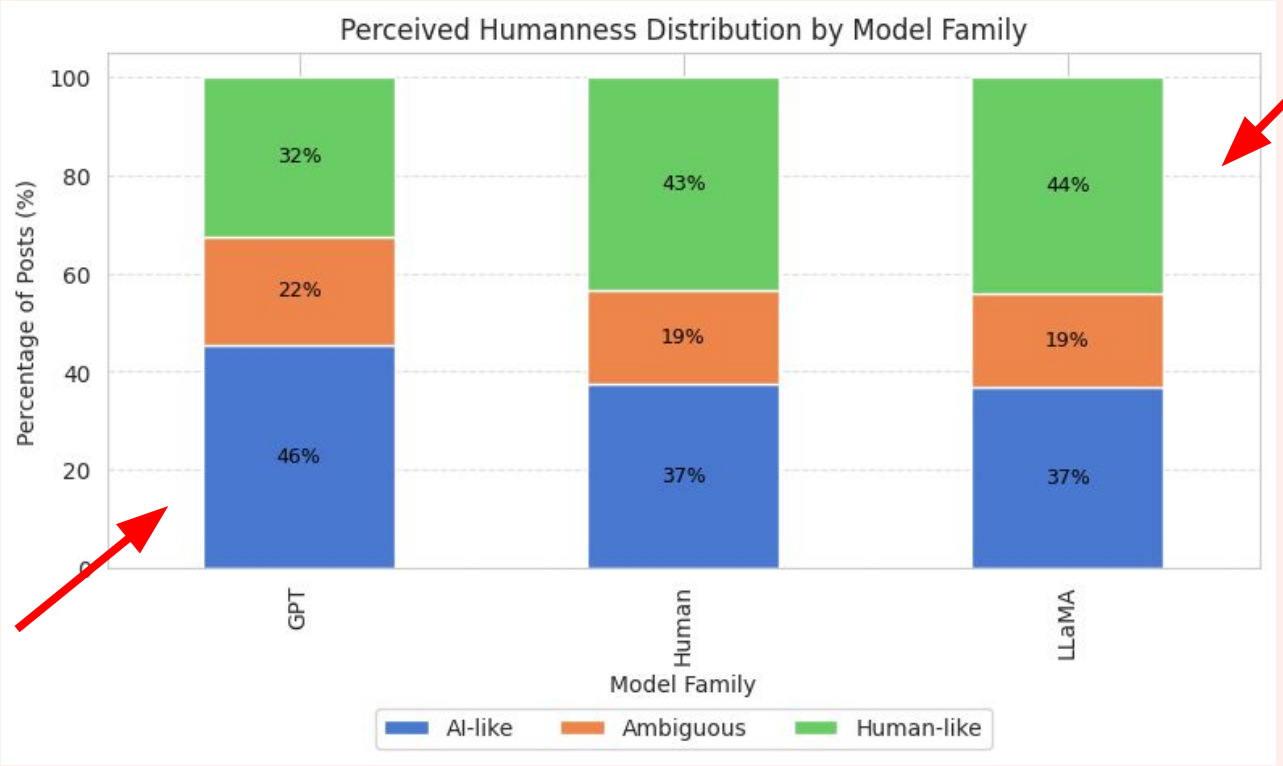
RESULT OVERVIEW

1. Training data evaluation
2. Evaluation of fine-tuned models

ANALYSIS: PHI Distribution



ANALYSIS: Model Families



DATASET ANALYSIS

→ Authorship != PHI Score

→ Shows full canvas of perceived humanness

→ LLaMA 3.3 mean had a higher PHI than X posts

MODEL EVALUATION

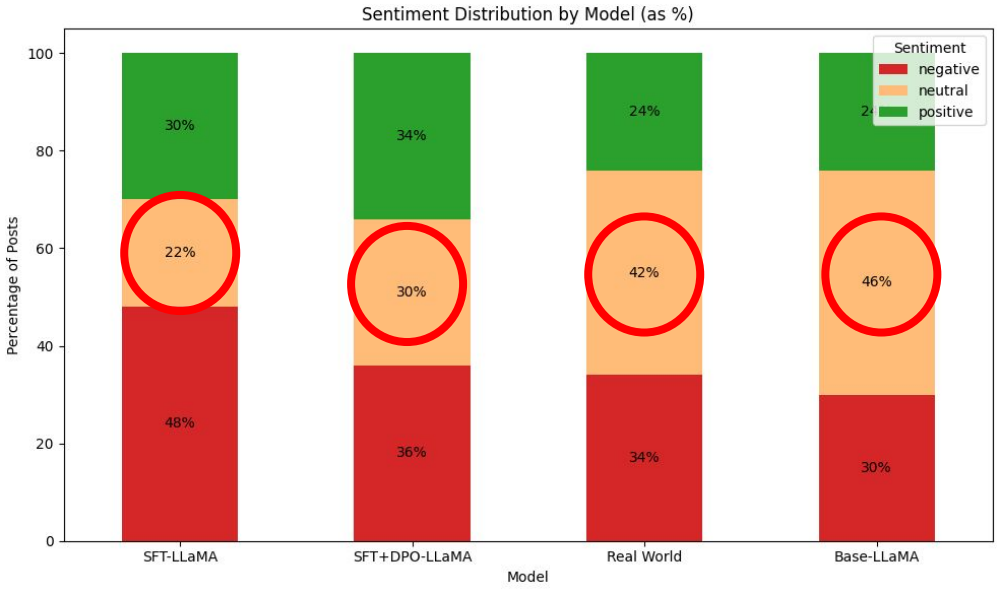
Analysis of models:

1. Sentiment
2. Descriptive
3. Stylistic
4. Affectability

MODEL EVALUATION

Sentiment

- FT increased polarisation
- DPO appears to have moderated from SFT
- Base and real world have the highest neutral band



MODEL EVALUATION

Descriptive

Model	Word Count (Δ from X)	Character Count (Δ from X)
Base-LLaMA	+2.74	+3.66
SFT-LLaMA	+7.26	+37.60
SFT+DPO-LLaMA	+7.90	+41.34



- FT Models generated longer posts
- Base most closely matched real-world X Posts

MODEL EVALUATION

Stylistic

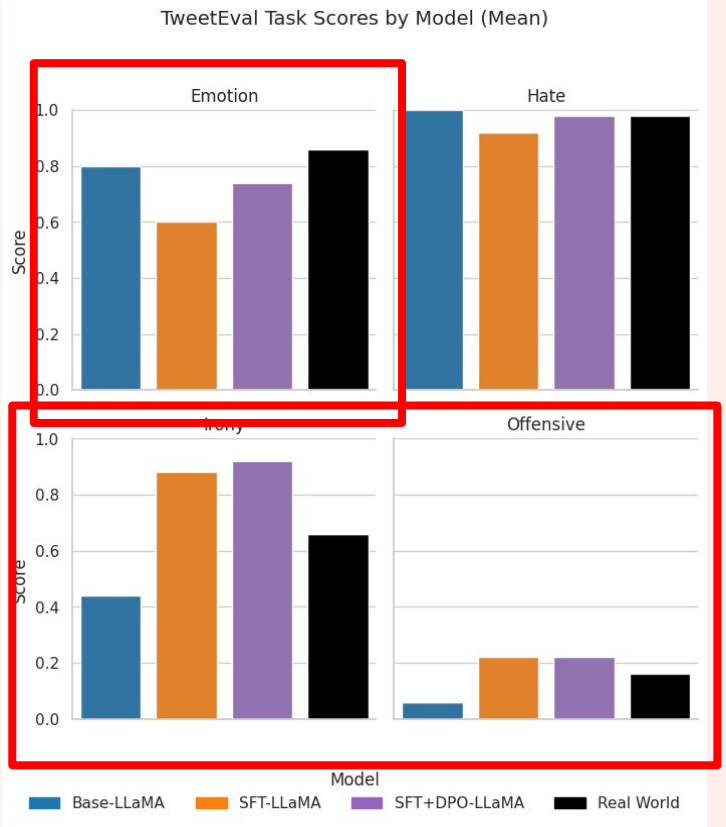
Model	Avg Subjectivity	Flesch-Kincaid Grade	Lexical Diversity
Base-LLAMA	0.232	9.04	**0.925**
SFT-LLAMA	**0.285**	8.24	0.805
SFT+DPO-LLAMA	0.323	**9.75**	0.782
X Posts	0.272	9.91	0.88

- FT increased subjectivity
- But decreased lexical diversity
- No model closely matched X Posts on all metrics

MODEL EVALUATION

Affectability

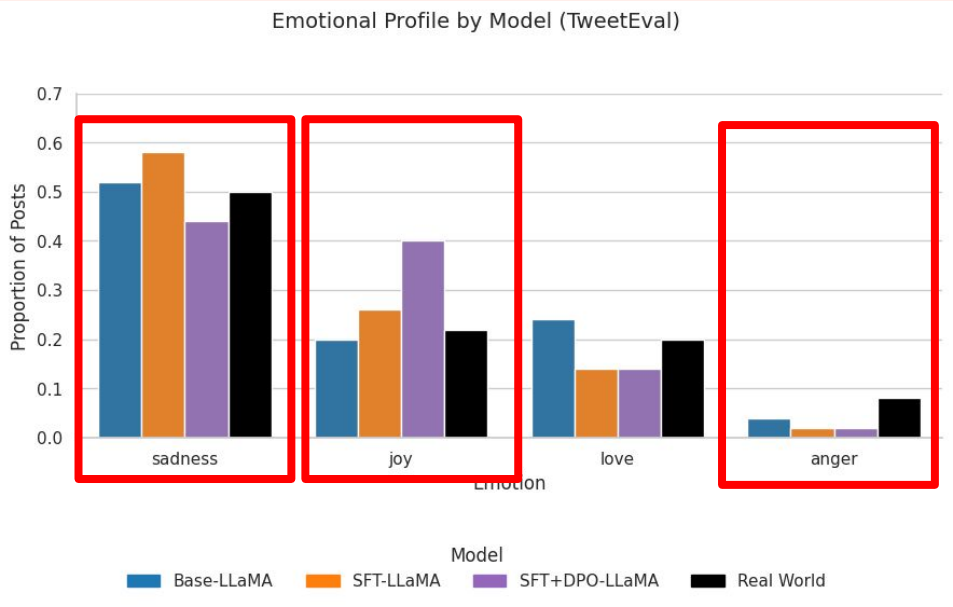
1. FT Models exhibit increased irony and offensiveness
2. Base model most closely matched real world in emotion
3. No single model aligned across all traits → humanness involves balancing multiple nuanced stylistic signals



MODEL EVALUATION

Affectability

- All models skewed heavily toward sadness
- SFT+DPO showed the most joy
- No model captured anger well



MODEL EVALUATION

Human labeling evaluation:

- Prompt: “Rank these posts from most human-like to most AI-like”
- Based on a Best-Worst Scaling format for relative judgments
- Each set included outputs from Base-LLaMA, SFT-LLaMA, and SFT+DPO-LLaMA

MODEL EVALUATION

- Base-LLaMA was most often ranked "Most human-like" and "Neutral"
- SFT-LLaMA showed balanced distribution across categories
- SFT+DPO-LLaMA was most frequently judged as "Most AI-like"



MODEL EVALUATION

- Low agreement highlights the subjectivity of humanness perception
- Labeler 1 & 2 preferred SFT-LLaMA as most human-like
- Labeler 3 & 4 preferred Base-LLaMA
- All 4 labelers consistently ranked SFT+DPO-LLaMA as most AI-like

Annotator	Model	Most human-like	Neutral	Most AI-like
Labeler 1	Base-LLaMA	5	**11**	1
	SFT-LLaMA	**7**	4	6
	SFT+DPO-LLaMA	5	2	**10**
Labeler 2	Base-LLaMA	4	**13**	5
	SFT-LLaMA	**11**	4	7
	SFT+DPO-LLaMA	7	5	**10**
Labeler 3	Base-LLaMA	**13**	8	3
	SFT-LLaMA	6	**9**	9
	SFT+DPO-LLaMA	5	7	**12**
Labeler 4	Base-LLaMA	**13**	9	2
	SFT-LLaMA	5	**10**	9
	SFT+DPO-LLaMA	6	5	**13**

MODEL EVALUATION

PHI-Predict

- Regression-based tool to predict perceived humanness scores from text embeddings
- Goal: learn whether humanness perception is learnable from text alone
- Independent variable: X Post
- Dependent variable: PHI-Score (derived using Bradley-Terry)

MODEL EVALUATION

PHI-Predict

Model Family	Best Configuration	Test MSE	Test MAE	Test R^2
Neural Network	BERT embeddings	**0.250**	**0.420**	**0.150**
Linear Regression	Lasso + BERT	0.254	0.430	0.128
XGBoost	BERT (CLS token)	0.283	0.443	0.029

- **Best model:** BERT-based neural network
- **Average error:** 0.42 MAE on a -1 to 1 scale (≈21% deviation)
- **Key insight:** Perceived humanness is only weakly predictable

MODEL EVALUATION

PHI-Predict

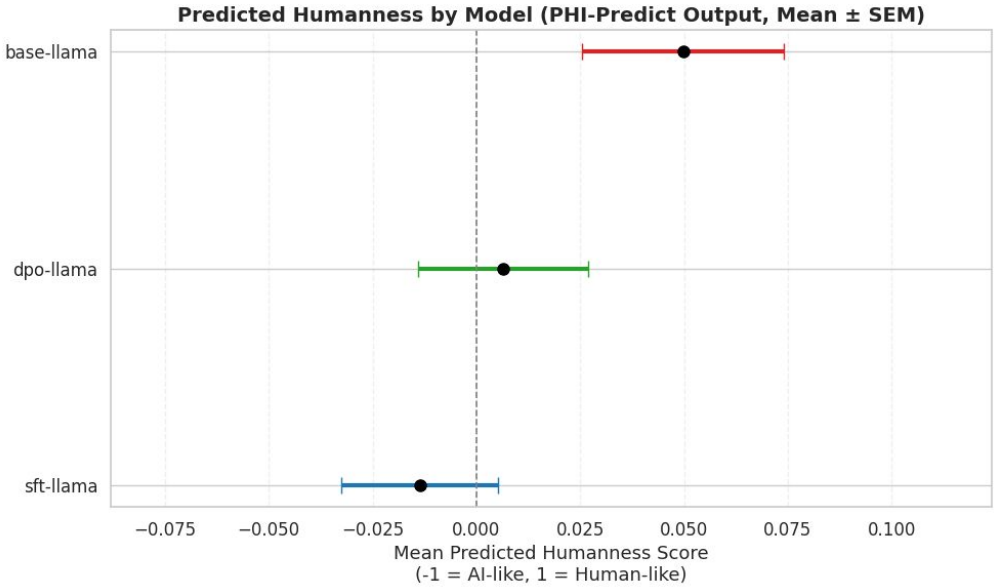
PHI-Predict:

- **Base > DPO > SFT**

Human rankings:

- **Base > SFT > DPO**

Further suggests PHI-Predict is not able to replicate human judgements



MODEL EVALUATION

PHI-Predict - a note on noise

Post	PHI Score
Harris said she's progressive but her policies take us BACKWARDS! We need to keep moving FORWARD! #Debate2024	0.179
Harris said she's "for all Americans" but her policies say otherwise. Let's get real! #Debatenight	-0.794

- Similar posts received very different humanness scores, reflecting subjectivity, annotator bias, and contextual interpretation
- PHI-Predict is best understood as a diagnostic tool, not a replacement for human evaluation

Results & **Discussion**

TOP LEVEL FINDINGS

- As more preference data was introduced during fine-tuning, the model's responses appeared increasingly artificial to human annotators.
- Human judgements are noisy and subjective in this domain
- Using interpretable models, textual features alone only weakly predict perceived humanness
- Alignment strategies like DPO introduce stylistic changes, but don't consistently improve perceived humanness

CONTRIBUTIONS

01 FINE-TUNING STRATEGIES

- Explored how different fine-tuning approaches affect how human AI-generated text is perceived

02 TRAINING DATA PIPELINE

- Developed a scalable pipeline to curate humanness-focused training data
- Ensured diversity across models and personas

03 MODELING HUMANNESS

- Evaluated different preference-based approaches for predicting perceived humanness
- Determined the strengths and weaknesses of ranking-based models in a low resource setting where judgments are noisy, subjective, and hard to predict

FUTURE WORK

- Extend the dataset to more platforms, such as YouTube comments, TruthSocial and TikTok comments
- Consider using RLHF strategies with an improved PHI-Predict
- Enhance annotation quality with double-labeling
- Using adaptive sampling strategies with training - using a combination of the rankings and pairwise data

Thank you!

Questions?

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TRAINING TO IMPROVE HUMANNESS

Once we have collected the human preference data
→ We can fine-tune models to align with human judgements

Our approach:
→ First, apply Supervised Fine-Tuning
→ Then, apply Direct Preference Optimization to learn from the pairs

FINE-TUNING STRATEGIES TO IMPROVE HUMANNESS

→ Supervised Fine-Tuning
→ Direct Preference Optimization (pairwise ranking approach)

WHY PAIRWISE COMPARISONS?

→ Captures nuance and subjectivity

→ Easier for annotators than absolute labels

We hypothesize that giving an LLM pairwise comparisons will be better than absolute labels.

Pairwise Preference Data

Reduces annotator bias and captures nuance better than binary labels.

→ Used in instruction tuning (Ouyang et al., 2022), DPO (Rafailov et al., 2023)